MASOUD ABDOLLAHI, Ph.D.

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PROFESSIONAL SUMMARY

AI/ML Research Scientist with a Ph.D. and 4+ years building and shipping production AI systems in high-stakes, data-rich environments: healthcare and industrial settings where accuracy and reliability are non-negotiable. Deep hands-on expertise in large language models, vision-language models, generative AI, self-supervised learning, and multimodal deep learning on time-series and imaging data. Track record of closing the loop from fundamental research to production deployment, with **28+ peer-reviewed publications, 700+ citations, and 6 patent filings (1 published).** Currently applying clinical AI models for neurological impairment detection at Oregon Health & Science University.

CORE TECHNICAL EXPERTISE

- **Generative & Foundation Models:** LLMs, VLMs, prompt tuning / prompt engineering, fine-tuning, distillation, quantization, RAG
- **Deep Learning Architectures:** Transformers, CNN, RNN-LSTM, semi-supervised & self-supervised learning, transfer learning, multimodal fusion
- **Healthcare & Clinical Data:** Time-series waveform analysis (IMU, clinical sensors), electronic medical records, clinical report processing, fall risk and neurological assessment models
- **Frameworks & Tools:** PyTorch, TensorFlow, Hugging Face Transformers, Azure OpenAI Foundry, Azure ML, OpenCV, MATLAB, SQL, Python, C++
- **Responsible AI & MLOps:** Robustness evaluation, statistical error estimation, end-to-end ML pipelines, production deployment, large-scale dataset design and curation

PROFESSIONAL EXPERIENCE

Senior R&D Scientist · Hitachi America, Ltd., Novi, MI *Feb 2023 – Present*

- **Technical leadership:** Promoted to Senior R&D Scientist (Jun 2026); lead multiple R&D projects and own a research roadmap.
- **Generative AI for clinical-scale automation:** Led end-to-end research, training, and production deployment of a Vision-Language Model (VLM) + generative AI system that automated a complex bottleneck process, reducing cycle time by **25%** across a live manufacturing facility, prototype to full rollout.
- **LLM-powered automation of structured reports:** Designed and built Transformer-based models on structured OEM data to automate classification, anomaly detection, and quality assessment tasks directly analogous to clinical NLP pipelines over EMRs and clinical reports.
- **Multimodal learning on wearables + vision:** Developed multimodal perception systems fusing IMU wearable sensor streams and computer vision (pose estimation, action recognition) achieving **>90% classification accuracy** across diverse real-world users.
- **Responsible AI in practice:** Applied Azure OpenAI Foundry for model development with evaluation workflows aligned to responsible AI principles; conducted rigorous statistical analyses and error estimation to meet production accuracy requirements on diverse, real-world subject populations.
- **Large-scale real-world data pipelines:** Collected, curated, and managed 500+ hours/year of in-field deployment data; built robust training and evaluation pipelines from heterogeneous sensor and vision streams.
- **IP and cross-functional impact:** Filed 6 patents in AI/ML, generative AI, and sensing (inventor on all; 1 published, 2 under USPTO review, 3 approved for filing); presented results to leadership and external partners.
- **AI Data Challenge winner:** Won the 2026 Hitachi America R&D AI Data Challenge with a physics-informed, data-efficient model that forecasts day-ahead hourly electricity prices at all 243 buses of a power-grid benchmark from only 30 labeled days.

Research Engineer III (Part-time) · Oregon Health & Science University, Remote *Feb 2024 – Present*

- **Wearable-sensor AI for Parkinson's disease:** Developing sensor-based deep learning models to analyze freezing of gait and fall risk in Parkinson's disease, applying time-series ML and signal processing to

wearable-derived data in a research-to-deployment pipeline alongside clinicians and medical domain experts.
- Bridging quantitative ML modeling and clinical domain knowledge; experience informing responsible AI development for patient-centered applications.

R&D Scientist, Internship · Aster Pet Sciences *Jun 2021 – Aug 2021*
- Owned an AI project from inception to delivery: led data collection, model design, and shipment of a real-time IMU-based activity recognition system (SVM, >70% accuracy).

EDUCATION

Ph.D., Mechanical and Industrial Engineering · Rochester Institute of Technology 2017 – 2023
Dissertation: *Machine Learning for Fall Risk Assessment: integrating clinical tests, wearable motion sensors, and multimodal data. A direct precursor to building multimodal AI for healthcare.*

M.Sc., Mechanical Engineering · Sharif University of Technology 2012 – 2015

SELECTED PUBLICATIONS

28+ peer-reviewed publications · 700+ citations · Full list on Google Scholar

1. Abdollahi, M., Arcobelli, V. A., Mancini, M., Horak, F. B., et al. (2026). *360° turning in individuals with mild to moderate Parkinson's disease: a clustering analysis of segmental coordination strategies.* Journal of NeuroEngineering and Rehabilitation. [DOI]
2. Vidyaratne, L., Kumar, A., Zhou, Q., Abdollahi, M., et al. (2026). *Generative AI framework for procedural knowledge extraction and training content generation in manufacturing.* ASME MSEC 2026, V002T07A007. [DOI]
3. Abdollahi, M., Zhou, Q., & Yuan, W. (2024). *Smart wearable insoles in industrial environments: A systematic review.* Applied Ergonomics 118, 104250. [PDF]
4. Abdollahi, M., Rashedi, E., et al. (2024). *Fall risk assessment in stroke survivors: a machine learning model using detailed motion data.* Sensors 24(3), 812. [PDF]
5. Jahangiri, S., Abdollahi, M., et al. (2024). *An inpatient fall risk assessment tool: Application of ML models on intrinsic and extrinsic risk factors.* Machine Learning with Applications 15, 100519. [PDF]
6. Jahangiri, S., Abdollahi, M., et al. (2024). *A machine learning model to predict heart failure readmission: toward optimal feature set.* Frontiers in Artificial Intelligence 7, 1363226. [PDF]
7. Karvekar, S., Abdollahi, M., & Rashedi, E. (2021). *Smartphone-based human fatigue level detection using machine learning.* Ergonomics (2021): 1–28. [PDF]

PATENTS

- Zhou, Q., & Abdollahi, M. Smart Sole Architecture for Measuring Ground Reaction Force and Method for Unifying Coordinates for Feet Positions and Gestures. U.S. Patent Application Publication US 2026/0198641 A1. Published Jul. 16, 2026.
- Abdollahi, M., et al. Agentic AI & generative AI system for industrial operations. U.S. Patent Application. Under USPTO Review. Hitachi America, Ltd. (2025).
- Abdollahi, M., et al. Sensor-based human performance assessment system. U.S. Patent Application. Under USPTO Review. Hitachi America, Ltd. (2026).
- Abdollahi, M., et al. Agentic AI, generative AI, sensor-based human performance assessment & training (3 applications). U.S. Patent Applications. Approved for filing; in preparation with Hitachi America, Ltd. IP & Legal (2026).